Frequency-Guided Progressive Transformer for Change Detection in Remote Sensing Images

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***Abstract*—Change detection (CD) aims to delineate land-cover changes from bitemporal remote sensing images (RSIs), yet pseudo-changes induce widespread false alarms because of illumination variation and atmospheric effects frequently. Moreover, recent CD studies under diverse land-cover distributions often suffer from fragmented structures and blurred boundaries. To address these issues, we propose a frequency-guided progressive transformer (FGPFormer) for CD in RSIs, which leverages frequency-based transforms to disentangle the true and pseudo-changes and integrate a change-assisted feature to guide precise decoding. Specifically, a texture-oriented frequency attention (TFA) module attenuates global low-frequency pseudo-changes by adaptive frequency selection and enhances change perception with directional texture guidance via frequency cross-attention. Furthermore, a structure-aware frequency refinement (SFR) module aggregates multilevel features and employs a wavelet transform to promote regional coherence and sharpen boundaries. Besides, a lightweight change-assisted segmentor (CAS) propagates the learned frequency structure guidance across decoding stages to ensure cross-scale consistency for multiscaled detection. Extensive experiments on four public benchmarks demonstrate that FGPFormer substantially reduces shift-induced false positives and produces more boundary-complete change results than state-of-the-art (SOTA) methods. The code will be released at https://github.com/JwNing96/FGPFormer**

***Index Terms*—Boundary-aware decoding, Fourier transform, frequency domain learning, remote sensing change detection (RSCD), transformer, wavelet decomposition.**

I. Introduction

REMOTE sensing change detection (RSCD) aims to localize land use (LU) and land-cover (LC) changes from bitemporal observations [1], [2], and serves as a fundamental component in numerous earth observation applications, including LU planning [3], LC monitoring [4], postdisaster reconstruction [5], and resource management [6]. With the increasing availability of high-resolution satellite imagery, practical RSCD methods are expected to deliver accurate and boundary-complete change mapping under diverse imaging conditions. In real deployments, false alarms drastically increase manual verification costs and mislead downstream decisions, while missed detections overlook critical events such as illegal construction or disaster-induced damages [7]. Developing robust RSCD models that suppress pseudo-changes and precisely delineate true changes remains an important and challenging problem.

Recent years have witnessed substantial progress in RSCD driven by deep learning [7]. Most state-of-the-art (SOTA) approaches adopted Siamese-style encoders to extract bitemporal features and infer changes via fully convolutional network [8], [9], spatial-channel attention-based fusion [10], [11], and transformer-based interactions [12], [13]. Multiscale decoders are further employed to integrate high-level semantics and low-level details for different resolution RSCD [14], [15]. Despite these advances, performance in real-world scenarios often degrades significantly due to imaging-condition inconsistency and the difficulty of preserving structural details, especially when changes are of various scales and structurally complex.

As illustrated in Fig. 1(a), a major reason for performance degradation is pseudo-change regions caused by appearance shift [16]. Bitemporal remote sensing images (RSIs) are frequently captured under varying illumination, atmospheric scattering, sensor properties, radiometric correction pipelines, and viewing geometries. These factors induce radiometric and color shifts even when no change occurs, resulting in widespread false alarms. Moreover, existing spatial-domain fusion and attention mechanisms often amplify such global low-frequency variations, making models over-sensitive to nonchange variations and limiting their generalization across scenes and imaging conditions [1]. In contrast, as shown in Fig. 1(b), true changes are commonly characterized by boundary-aligned discontinuities and coherent textures, which correspond to high-frequency structural responses. Unfortunately, this guidance is unsatisfactorily weakened by downsampling-upsampling operations and mul-

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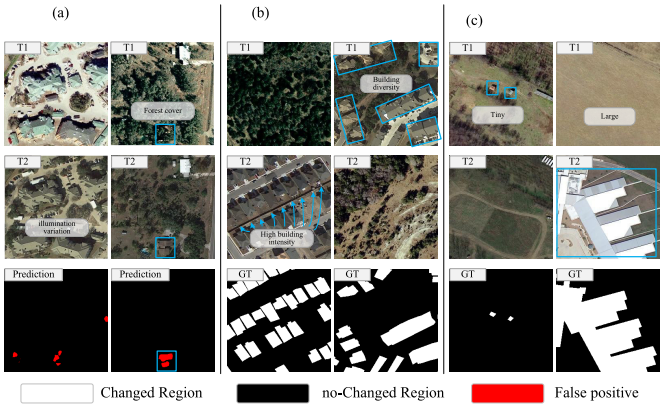


Fig. 1. Illustration of three major challenges in remote sensing CD. T1, T2, GT, and predict indicate prechanged image, postchanged image, changed label, and prediction, respectively. (a) *False Positives in Pseudo-Change Regions:* Appearance shift caused by illumination or atmosphere differences produces large contiguous false activations even without physical changes. (b) *Illegible Structures and Boundaries:* High-frequency structural discontinuities that indicate true changes are weakened by downsampling-upsampling and imperfect fusion, leading to blurry and incomplete change boundaries. (c) *Cross-Scale Inconsistency:* It lacks sensitivity in extensive changes and subtle variations of land features, leading to significant false negatives in RSCD across different object sizes.

tiscale fusion, leading to fragmented predictions caused by blurred boundaries and incomplete object shapes, particularly within obscured and shadowed regions [1]. Such degradation is particularly evident in high-resolution scenes with dense LU structures, where precise boundary delineation is critical. The problem becomes more severe for fine-grained changes and changes occurring on slender structures (e.g., narrow buildings and roads), where change evidence is often subtle and low-contrast, thus easily overwhelmed by dominant contextual features. Moreover, as shown in Fig. 1(c), multiscale inconsistency, which lacks sensitivity in extensive changes and subtle variations of land features, leads to significant false negatives in RSCD across different object sizes. These observations highlight that robust RSCD requires not only restraining pseudo-changes under low-frequency shift, but also maintaining the consistency of structure and scale. In summary, three issues are listed as follows.

1) Illumination and atmospheric discrepancies induce pseudo-change regions, yielding false positives and ambiguous activations even in the absence of actual physical changes.
2) Shadows and occlusions often blur true-change boundaries, resulting in incomplete detections and disrupted continuity of change structures.
3) Current RSCD methods are often sensitive to variations in change scale, which undermines prediction consistency and increases false negatives.

To tackle these challenges, we revisit RSCD from a frequency perspective and investigate an effective and efficient frequency–structure joint solution. Pseudo-changes are mainly attributable to low-frequency background shift, while true changes primarily manifest as high-frequency structural discontinuities. Therefore, fast Fourier transform (FFT) is introduced as a principled mechanism to disentangle change evidence from nonchange variations by attenuating low-frequency shift and enhancing the integrity of the high-frequency change. Nevertheless, FFT is global and lacks spatial locality, which is crucial for encoding fine-grained texture and structure. To bridge this gap, the discrete wavelet transform (DWT) provides a localized and fine-grained decomposition that serves as an explicit structure-aware prior. Its low-frequency component promotes region-level consistency, while its high-frequency component preserves boundary connectivity and sharpens edge delineation. Consequently, a joint frequency–structure solution that couples FFT-based global frequency decoupling with DWT-based multiscale structural refinement is particularly suited to coarse-to-fine RSCD.

Based on these insights, we propose a frequency-guided progressive transformer (FGPFormer) for change detection (CD) of RSIs. FGPFormer consists of three modules, including texture-oriented frequency attention (TFA), structure-aware frequency refinement (SFR), and change-assisted segmentor (CAS). Specifically, TFA performs FFT-based adaptive frequency selection on stage-wise bitemporal difference features to suppress low-frequency appearance shifts that usually induce pseudo-changes. To recover local structure and boundary details that are easily diluted in deep encoders, SFR further constructs a structured frequency guidance through DWT, which jointly stabilizes region-level predictions and sharpens change boundaries. CAS injects both TFA-derived change-aware frequency features and SFR-derived wavelet structure priors into the multiscale decoding process. This progressive guidance mechanism encourages cross-scale consistency and improves the detection of both large-area changes and fine-grained objects. Extensive experiments on four benchmarks demonstrate that FGPFormer yields more accurate and boundary-complete change maps, achieving notable gains on boundary-sensitive metrics and significantly reducing false alarms under illumination variations compared with recent SOTA baselines.

The main contributions of this work are summarized as follows.

1) We propose an FGPFormer for RSCD, which establishes an effective and efficient framework guided by two frequency perception and a CAS.
2) TFA module is presented to perform adaptive frequency selection and integrates directional texture guidance via a cross-attention mechanism, preventing false alarms under low-frequency background shift.
3) SFR module is constructed via multiscale feature aggregation and DWT, which explicates the complementary relationship between low-frequency region consistency and high-frequency boundary sharpening, refining the structure and boundary accuracy of the changed areas.
4) CAS injects change-aware features of TFA and wavelet structure priors of SFR into the multiscale decoder, achieving cross-scale consistent CD and enhancing the integrity of large-scale and fine-grained objects.

The rest of this article is organized as follows. Section II describes the existing related works. Section III elaborates on the details of the proposed FGPFormer. Experiments are demonstrated in Section IV. Finally, the conclusion of this article is summarized in Section V.

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II. Related Work

A. Deep-Learning-Based Remote Sensing CD

Deep learning has substantially advanced RSCD in recent years, largely by formulating the task as a dense prediction problem over bitemporal RSIs [1], [7]. A dominant paradigm is siamese learning, where two temporal images are encoded by weight-sharing backbones and changes are inferred from feature differences or cross-temporal interactions [8], [17]. Representative designs include difference-learning networks that explicitly model discrepancies between temporal features, as well as encoder-decoder architectures that aggregate multilevel representations to balance semantic discrimination and spatial detail [10], [18]. Multiscale decoding and feature pyramid strategies are also widely adopted to handle the scale variation of changed objects and to improve dense localization quality [19], [20]. More recently, multihead self- and cross-attention mechanisms in transformer-based architectures have become mainstream, as they enhance long-range dependency modeling and strengthen cross-temporal feature interaction for improved change reasoning [12], [13], [21], [22].

Regrading pseudo-changes caused by illumination variations and atmospheric scattering, existing efforts [9], [15], [23] improve robustness mainly from the spatial domain and the feature domain perspective. DDRL [23] constrained feature distributions via graph-enhanced representation learning and contrastive objectives, aiming to separate true-change features from pseudo-change features under domain shifts. IA-CDNet [15] targeted adverse imaging conditions by integrating an image-adaptive enhancement into the RSCD pipeline, improving robustness without redesigning the backbone. HFTN [9] suppressed noise-driven artifacts through progressive denoising together with hierarchical feature fusion, stabilizing change maps under noisy observations. These works reduce false alarms under appearance variations, but they lack an explicit model of pseudo-changes as low-frequency shift and thus remain susceptible to shift-induced activations in challenging scenes.

In terms of fragmented structures and degraded boundaries, structure- and boundary-aware methods [11], [24], [25] have been developed to mitigate this issue. SFEARNet [11] combined semantic flow transmission with edge-aware refinement to reduce detail loss during decoding and improve boundary sharpness. SEDANet [24] identified erroneous attention as a key failure mode and learns ensemble difference attention weights to better focus on foreground changes, improving building boundary integrity. Beyond refinement and attention correction, relational reasoning has been explored to enhance structural consistency. For instance, MAFGNet [25] employed multiscale attention fusion and dual graph convolutions to capture long-range dependencies and relational structures, benefiting structured changes. While these methods improve boundary completeness, they largely operate in the spatial and feature domain with excess data processing and structural guidance.

Moreover, multiscale decoding is essential for jointly handling large-area changes and fine-grained changes [26], [27]. In this case, change evidence may be activated at one scale but suppressed at another, yielding unstable predictions and scale-dependent artifacts [7]. In parallel, data-efficient learning is becoming increasingly important. STMNet [28] explored a single-temporal, mask-based self-supervised framework for hyperspectral CD, reducing reliance on paired bitemporal labels. Changen2 [29] adopted a generative foundation model approach by synthesizing multitemporal imagery and change supervision for scalable pretraining and stronger transferability. Despite these progresses, the key difficulties observed in practical RSCD largely align with the three issues emphasized in Section I, which motivates a unified mechanism that explicitly separates low-frequency appearance shift from high-frequency structural change guidance and propagates them in a coarse-to-fine manner.

B. Frequency-Guided Mechanisms

Frequency-domain modeling, which is widely applied in visual downstream tasks [30], [31], [32], provides a complementary perspective to spatial-domain processing by explicitly characterizing low-frequency global variations and high-frequency structural details [32]. Specifically, FADC [30] related dilated convolution to local spectra and predicted spatially adaptive dilation with frequency-aware kernel decomposition and selection to balance effective bandwidth and receptive field. SegCFT [31] replaced self-attention with a context-aware Fourier transform and utilized hierarchical frequency mixing with adaptive modulation to align frequency cues with spatial features for efficient dense prediction. FreqFusion [32] interpreted feature-fusion failures as distorted or missing high-frequency details and improved fusion via adaptive low-pass filtering, offset-guided resampling, and high-pass enhancement for sharper boundaries.

In RS scenarios, it greatly inspires the model to distinguish the foreground from the background and highlight the region of interest by utilizing a frequency-guided mechanism. Recent studies exploited explicit frequency decomposition to complement spatial representations in RS tasks. For instance, SFT [33] learned frequency-space attention to filter spectral components and enforced class-activation-guided consistency to improve sample reliability for hyperspectral cross-scene classification. In the RS segmentation task, representative lines of work included dual-domain transformers that coupled spatial modeling with an FFT branch and learnable spectral masking under boundary-aware supervision [34], wavelet fusion networks that injected Haar subbands and aligned spatial and frequency feature via multihead cross-attention [35], and frequency decoupling or enhancement frameworks that employed DCT [36] or FFT [37] masks to separately refine low-frequency context and high-frequency details, often together with global–local integration in Swin-style backbones [38]. From a frequency-domain perspective, these studies suggested that mitigating latent low-frequency shifts while enhancing salient high-frequency structures yields more robust and boundary-accurate predictions.

For RSCD, recent studies increasingly introduced explicit frequency-domain priors to mitigate imaging-condition shift and to sharpen structural boundaries. For instance, FIMP [16] mainly filters bitemporal features in the Fourier domain to

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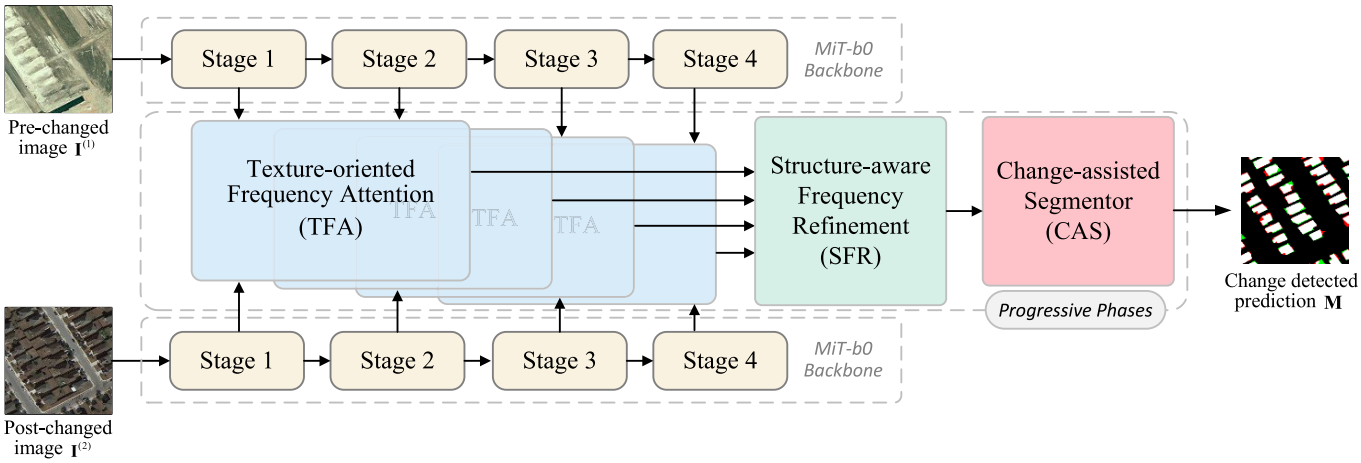


Fig. 2. Overview of the proposed FGPFormer for remote sensing CD. A siamese MiT-b0 encoder extracts a four-stage feature pyramid from the prechange image $\mathrm{I}^{(1)}$ and the postchange image $\mathrm{I}^{(2)}$. TFA suppresses shift-induced pseudo-changes in the low-frequency spectrum while enhancing structure-relevant responses. SFR further aggregates multilevel cues with wavelet-based guidance to improve region consistency and boundary sharpness. A lightweight CAS integrates the refined features and outputs the final change prediction.

reduce style-driven pseudo-changes, but it does not introduce localized wavelet-based structural refinement or decoder-level progressive guidance. FDINet [39] employs wavelet decomposition for band-specific interactions, but it does not explicitly model low-frequency pseudo-change suppression through adaptive Fourier selection. EFICNN [40] enhances high-frequency boundaries using Laplacian-pyramid edge attention and flow-guided fusion, while it does not jointly formulate low-frequency shift suppression, wavelet-based structure refinement, and cross-scale decoding consistency within a unified RSCD framework. In contrast, FGPFormer establishes a progressive pipeline of frequency decoupling, structure refinement, and change-assisted decoding, where TFA, SFR, and CAS are functionally complementary.

III. Methodology

A. Problem Definition

Given a bitemporal image pair $\mathbf{I}^{(1)}, \mathbf{I}^{(2)} \in \mathbb{R}^{H\times W\times C}$, where H and W denote spatial height and width, and C denotes the number of channels, we predict a binary change mask $\mathbf{Y} \in \{0,1\}^{H\times W}$, where $Y_{uv} = 1$ indicates a changed pixel at location (u, v). Our network is a parametric function $f_\Theta(\cdot)$ with learnable parameters Θ. It outputs a per-pixel logit tensor $\mathbf{M} \in \mathbb{R}^{H\times W\times 2}$ for two classes (unchanged and changed), and a posterior probability map $\hat{\mathbf{Y}} \in [0,1]^{H\times W}$ computed by a per-pixel softmax along the class dimension

$$\mathbf{M} = f_\Theta\left(\mathbf{I}^{(1)}, \mathbf{I}^{(2)}\right), \quad \hat{\mathbf{Y}}_{uv} = \mathrm{Softmax}\left(\mathbf{M}_{uv}\right)_1 \tag{1}$$

where $\mathbf{M}_{uv} \in \mathbb{R}^2$ is the logit vector at pixel (u, v) and the subscript $_1$ denotes the changed class.

B. Overview

We design a progressive pipeline consisting of TFA, SFR, and a CAS, as shown in Fig. 2. A Siamese MiT-b0 encoder with shared parameters extracts a four-stage feature pyramid for each input. Let $l \in \{1,2,3,4\}$ index the encoder stages from shallow to deep. We denote the stage-l feature of $t \in \{1,2\}$ by $\mathbf{F}_l^{(t)} \in \mathbb{R}^{C_l\times H_l\times W_l}$, where C_l is the channel dimension and (H_l, W_l) is the spatial resolution at stage l.

Instead of directly fusing bitemporal streams, we encourage an explicit stage-wise change representation by feature differencing at first

$$\Delta\mathbf{F}_l = \mathbf{F}_l^{(2)} - \mathbf{F}_l^{(1)} \tag{2}$$

where the subtraction is element-wise. TFA refines each $\Delta\mathbf{F}_l$ to produce a change-enhanced feature $\mathbf{Z}_l \in \mathbb{R}^{C_l\times H_l\times W_l}$. SFR aggregates $\{\mathbf{Z}_l\}_{l=1}^4$ into a structure-aware frequency guidance map $\mathbf{G} \in \mathbb{R}^{D\times H_1\times W_1}$, where D is a decoder embedding dimension and (H_1, W_1) is the highest feature resolution. Finally, CAS decodes $\{\mathbf{Z}_l\}$ under the guidance of $\mathbf{G}$ and outputs the final logits $\mathbf{M}$.

C. Texture-Oriented Frequency Attention

TFA is designed to attenuate pseudo-change activations while enhancing true-change evidence. Pseudo-changes are often dominated by low-frequency appearance shift, while true changes tend to introduce boundary-aligned high-frequency discontinuities and coherent textures. For each stage l, TFA maps the input change feature $\mathbf{X}_l = \Delta\mathbf{F}_l$ to the refined feature

$$\mathbf{Z}_l = \mathrm{TFA}_l\left(\mathbf{X}_l\right). \tag{3}$$

For clarity, we describe a generic stage and omit the subscript l below, writing the input as $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$.

1) Adaptive Frequency Selection: We use global statistics of $\mathbf{X}$ to infer an adaptive low-frequency window. Let GAP$(\cdot)$ denote global average pooling over spatial dimensions. The channel descriptor $\mathbf{g} \in \mathbb{R}^C$ is computed by $\mathbf{g} = \mathrm{GAP}(\mathbf{X})$. A lightweight predictor $\phi(\cdot)$ outputs two scalars in $(0, 1)$

$$[s_h, s_w] = \sigma\left(\phi\left(\mathbf{g}\right)\right) \tag{4}$$

where s_h and s_w control the retained low-frequency extent along height and width, and $\sigma(\cdot)$ is the sigmoid function.

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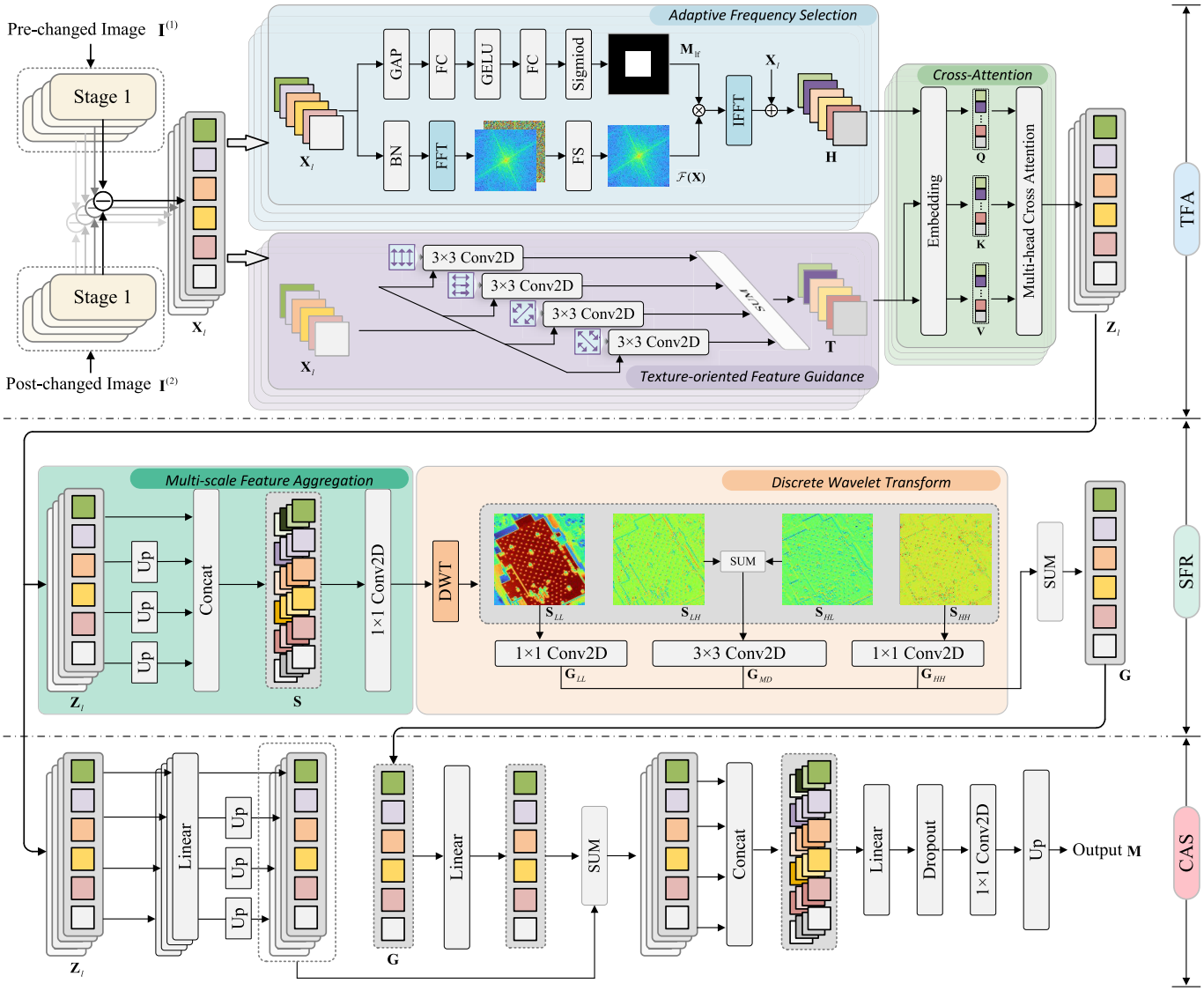


Fig. 3. Detailed architecture of the three progressive phases of the proposed FGPFormer.

We construct a binary low-frequency mask $\mathbf{M}_{\text{lf}} \in \{0,1\}^{H\times W}$ centered at the spectrum origin, whose window size is determined by (s_h, s_w). Let $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ denote the 2-D discrete Fourier transform and its inverse applied independently to each channel. Let $\odot$ denote element-wise multiplication. Thus, an adaptive high-pass operator $\mathcal{P}_{\text{H}}(\cdot)$ is defined as

$$\mathcal{P}_{\text{H}}(\mathbf{X};\mathbf{M}_{\text{lf}}) = \left|\mathcal{F}^{-1}\left(\mathcal{F}(\mathbf{X})\odot(1-\mathbf{M}_{\text{lf}})\right)\right| \tag{5}$$

where $|\cdot|$ denotes magnitude. The resulting frequency response is $\mathbf{H} = \mathcal{P}_{\text{H}}(\mathbf{X};\mathbf{M}_{\text{lf}}) \in \mathbb{R}^{C\times H\times W}$. This step alleviates low-frequency shift and preserves boundary-sensitive high-frequency evidence.

2) Texture-Oriented Feature Guidance: To model oriented textures that are common in aerial scenes, as shown in Fig. 3, we introduce a direction-aware convolutional operator. It produces four directional responses $\{\mathbf{T}^{(m)}\}_{m=1}^{4}$, where each $\mathbf{T}^{(m)} \in \mathbb{R}^{C\times H\times W}$ corresponds to one direction. We aggregate them using learnable channel-wise direction weights $\mathbf{W} \in \mathbb{R}^{C\times 4}$

$$\mathbf{T} = \sum_{m=1}^{4} W_m \odot \mathbf{T}^{(m)} \tag{6}$$

where $W_m \in \mathbb{R}^{C}$ denotes the weights for direction m. The aggregated texture feature $\mathbf{T} \in \mathbb{R}^{C\times H\times W}$ provides structure-aligned cues that help maintain boundary connectivity.

3) Cross-Attention Fusion: $\mathbf{H}$ and $\mathbf{T}$ are fused via multi-head cross-attention, where frequency features query texture features. Let $\psi_q(\cdot)$, $\psi_k(\cdot)$, and $\psi_v(\cdot)$ be learnable linear projections that map input features to queries, keys, and values. We compute

$$\mathbf{Q} = \psi_q(\mathbf{H}), \quad \mathbf{K} = \psi_k(\mathbf{T}), \quad \mathbf{V} = \psi_v(\mathbf{T}) \tag{7}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ share the same spatial resolution as $\mathbf{H}$ and $\mathbf{T}$. Then, we reshape them into token matrices of size $N \times d$, where $N = HW$ is the number of spatial tokens and d is the projected feature dimension. Let Norm$(\cdot)$ denote channel-

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wise vector normalization applied to each token. Therefore, normalized tokens are represented as

$$\tilde{\mathbf{Q}} = \mathrm{Norm}(\mathbf{Q}), \quad \tilde{\mathbf{K}} = \mathrm{Norm}(\mathbf{K}) \tag{8}$$

and compute attention using a learnable temperature parameter $\tau \in \mathbb{R}$

$$\begin{aligned} \mathbf{A} &= \mathrm{Softmax}\left(\tau \tilde{\mathbf{Q}} \tilde{\mathbf{K}}^{\top}\right) \\ \mathbf{O} &= \mathbf{A}\mathbf{V} \end{aligned} \tag{9}$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the attention weight matrix and $\mathbf{O} \in \mathbb{R}^{N \times d}$ is the attended output. A projection $\pi(\cdot)$ maps $\mathbf{O}$ back to the original channel dimension C and restores the spatial layout. The final TFA output is produced by a residual connection

$$\mathrm{TFA}(\mathbf{X}) = \mathbf{X} + \pi(\mathbf{O}). \tag{10}$$

This fusion transfers structure consistency from $\mathbf{T}$ to $\mathbf{H}$, which improves boundary completeness and suppresses fragmented pseudo-change responses.

D. Structure-Aware Frequency Refinement

TFA produces change-sensitive features at each stage, but decoding still requires balancing boundary sharpness and region-level stability. SFR constructs a structure-aware frequency prior utilizing localized wavelet decomposition, so that low-frequency components can stabilize regional decisions while mid- and high-frequency components reinforce connected boundaries and fine edges.

1) Multiscale Feature Aggregation: We align all refined stage features to the highest resolution and then aggregate them. As illustrated in Fig. 3, let $\mathrm{Up}(\cdot)$ denote bilinear upsampling to the stage-1 spatial size (H_1, W_1). We define aligned features $\tilde{\mathbf{Z}}_l \in \mathbb{R}^{C_l \times H_1 \times W_1}$ by

$$\tilde{\mathbf{Z}}_l = \begin{cases} \mathbf{Z}_1, & l = 1 \\ \mathrm{Up}(\mathbf{Z}_l), & l \in \{2,3,4\} \end{cases} \tag{11}$$

and aggregate them using a 1×1 convolution $\mathrm{Conv}_{1\times1}(\cdot)$

$$\mathbf{S} = \mathrm{Conv}_{1\times1}\left(\mathrm{Concat}_{l=1}^{4} \tilde{\mathbf{Z}}_l\right). \tag{12}$$

Here, $\mathrm{Concat}(\cdot)$ concatenates features along the channel dimension, and $\mathbf{S} \in \mathbb{R}^{C_s \times H_1 \times W_1}$ is the aggregated feature with channel dimension C_s.

2) Discrete Wavelet Transform: Let $\mathrm{DWT}(\cdot)$ denote a Haar DWT applied channel-wise. $\mathbf{S}$ is decomposed by $\mathrm{DWT}(\cdot)$ into four subbands

$$[\mathbf{S}_{LL}, \mathbf{S}_{LH}, \mathbf{S}_{HL}, \mathbf{S}_{HH}] = \mathrm{DWT}(\mathbf{S}) \tag{13}$$

where $\mathbf{S}_{LL}$ is the low-frequency subband, $\mathbf{S}_{LH}$ and $\mathbf{S}_{HL}$ are directional mid-frequency subbands, and $\mathbf{S}_{HH}$ is the high-frequency subband. Notably, all subbands share the same channel dimension as $\mathbf{S}$ and have half spatial resolution due to wavelet downsampling.

3) Frequency-Guided Structure-Aware Refinement: We refine the subbands utilizing three learnable convolutional layers

$$\begin{aligned} \mathbf{G}_{LL} &= \mathrm{Conv2d}_{1\times1}(\mathbf{S}_{LL}) \\ \mathbf{G}_{MD} &= \mathrm{Conv2d}_{3\times3}(\mathbf{S}_{LH} + \mathbf{S}_{HL}) \\ \mathbf{G}_{HH} &= \mathrm{Conv2d}_{1\times1}(\mathbf{S}_{HH}) \end{aligned} \tag{14}$$

where the addition $\mathbf{S}_{LH} + \mathbf{S}_{HL}$ is element-wise. Three flows are aggregated and then upsample back to (H_1, W_1) to form the guidance map

$$\mathbf{G} = \mathrm{Up}(\mathbf{G}_{LL} + \mathbf{G}_{MD} + \mathbf{G}_{HH}). \tag{15}$$

Finally, $\mathbf{G}$ is lifted to the decoder embedding dimension D using a pointwise projection so that $\mathbf{G} \in \mathbb{R}^{D \times H_1 \times W_1}$. It is obvious that $\mathbf{G}$ serves as a structure-aware frequency prior that discourages spatially incoherent pseudo-changes and reinforces boundary-consistent true changes.

E. Change-Assisted Segmentor

CAS is a lightweight segmentation head that aggregates multiscale evidence under the guidance of $\mathbf{G}$. It is designed to simultaneously handle thin and small changes requiring high-resolution details, as well as large-area changes requiring semantic consistency and regional stability.

For each stage l, the feature map $\mathbf{Z}_l$ is first projected into a unified embedding space

$$\mathbf{E}_l = \mathrm{Reshape}(\mathrm{MLP}_l(\mathbf{Z}_l)) \in \mathbb{R}^{D \times H_l \times W_l} \tag{16}$$

where $\mathrm{MLP}_l(\cdot)$ denotes a stage-specific linear embedding and $\mathrm{Reshape}(\cdot)$ restores the spatial layout.

Subsequently, all feature maps are upsampled to the highest decoding resolution (H_1, W_1) and conditioned by the SFR guidance feature through additive modulation

$$\hat{\mathbf{E}}_l = \mathrm{Up}(\mathbf{E}_l) + \mathrm{MLP}(\mathbf{G}), \quad l \in \{1,2,3,4\} \tag{17}$$

where $\hat{\mathbf{E}}_l \in \mathbb{R}^{D \times H_1 \times W_1}$ denotes the guidance-conditioned representation. This conditioning promotes cross-scale structural consistency under a shared prior, thereby improving boundary completeness and suppressing scattered pseudo-change responses.

The conditioned multiscale features are then concatenated and fused into a unified representation

$$\mathbf{F} = \mathrm{Fuse}\left(\mathrm{Concat}_{l=1}^{4} \hat{\mathbf{E}}_l\right) \tag{18}$$

where $\mathrm{Fuse}(\cdot)$ denotes a lightweight fusion block composed of pointwise convolution and nonlinear transformation.

Finally, the fused feature $\mathbf{F}$ is fed into a lightweight prediction head to generate the change logits $\mathbf{M}_1 \in \mathbb{R}^{H_1 \times W_1 \times 2}$. The prediction is then bilinearly upsampled to the original image resolution to obtain $\mathbf{M} \in \mathbb{R}^{H \times W \times 2}$, which is converted into the final probability map $\hat{\mathbf{Y}}$ using the softmax operation defined in Section III-A.

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F. Optimization Objective

We adopt pixel-wise cross-entropy for binary change segmentation. Let $\Omega = \{(u, v) \mid 1 \le u \le H, 1 \le v \le W\}$ denote the set of all pixel coordinates. Let $p_{uv}(c)$ be the predicted probability of class $c \in \{0, 1\}$ at pixel (u, v), defined as $p_{uv}(c) = \text{Softmax}(\mathbf{M}_{uv})_c$. Thus, the loss is formulated as

$$\mathcal{L}_{\text{ce}} = -\frac{1}{|\Omega|} \sum_{(u,v)\in\Omega} Y_{uv} \log p_{uv} \tag{19}$$

where $|\Omega| = HW$ is the number of pixels. The main objective is to minimize $\mathcal{L}_{\text{ce}}$ through network optimization.

IV. Experiments

A. Datasets

In our work, four public RSCD datasets, including LEVIR-CD, WHU-CD, DSIFN-CD, and CLCD, are employed to verify the effectiveness of the proposed FGPFomrer in various environments.

1) LEVIR-CD [41]: The dataset contains 637 very high-resolution image patch pairs with a size of 1024 × 1024 pixels from Google Earth API. These bitemporal images are from 20 different regions that sit in several cities in Texas, USA, with a ground sample distance (GSD) of 0.5 m. The capture time of our LEVIR-CD dataset varied from 2002 to 2018, with a time span of 5–14 years. On this dataset, seasonal changes and illumination changes are introduced to evaluate the effectiveness of the method that mitigates the impact of irrelevant changes on real changes.

2) WHU-CD [42]: The dataset consists of aerial and satellite datasets for building CD. In our experiments, we utilize its aerial sets, which cover most of the building area in the middle of the aerial dataset. It is seamlessly cropped into 8189 512 × 512 pairs with a GSD of 0.3 m. WHU-CD dataset contains countryside, residential, cultural, and industrial areas in Christchurch, New Zealand. Various and versatile architectural types of buildings with different colors, sizes, and usage make it an ideal study area to evaluate the potential of an RSCD algorithm.

3) DSIFN-CD [43]: It is a challenging dataset that is manually collected from Google Earth and consists of 394 image pairs for training and evaluating with a size of 512 × 512, covering Beijing, Chengdu, Shenzhen, Chongqing, and Wuhan. The Xi'an image pair is clipped into 48 image pairs for model testing. RSI acquired from an alternative urban setting is employed for model validation, with the objective of evaluating the generalizability of the models under out-of-distribution scenarios.

4) CLCD [44]: The dataset consists of 600 pairs of images of cropland change samples with a GSD ranging from 0.5–2 m. The bitemporal images in CLCD were collected by Gaofen-2 in Guangdong Province, China, in 2017 and 2019, respectively. Each image pair is composed of two images of 512×512 and a manually annotated binary label of cropland change. The main types of change annotated in the CLCD dataset include buildings, roads, lakes, and bare lands, which adequately assess the capacity of CD under the category diversity.

B. Implementation Details

1) Experimental Setup: The proposed FGPFormer employs a pretrained MiT-B0 backbone initialized on the ImageNet-1K dataset and is implemented using PyTorch 2.1.0 on Ubuntu 20.04. AdamW is used as the optimizer to optimize the model parameters, with a weight decay of 1×10^{-3} and an initial learning rate of 1×10^{-5}, and batch size and epoch are set as 8 and 300, respectively. The whole training process is on a single NVIDIA GeForce RTX 4090 GPU with 24 GB of memory. All the experimental setups for the comparison methods are in accordance with their original settings.

2) Data Preprocessing: Four utilized datasets are cropped to 256 × 256 pixels in a nonoverlapping manner, and the data augmentation leveraged includes flipping, rotating, and color jitter. It is noted that the DSIFN-CD dataset was repartitioned into a more rational data distribution to verify the comprehensive performance of the algorithms in generalization and veracity. Specifically, the validation set and the test set were first merged, and the combined dataset was then subjected to random and uniform partitioning.

3) Evaluation Metrics: In order to evaluate the performance of the proposed FGPFormer, we employ five evaluation metrics: overall accuracy (OA), precision (P), recall (R), $F1$ score ($F1$), and intersection over union (IoU). The definitions of the above metrics are as follows:

$$\text{P} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{20}$$

$$\text{R} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{21}$$

$$\text{F1} = \frac{2 \cdot \text{P} \cdot \text{R}}{\text{P} + \text{R}} \tag{22}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{23}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{24}$$

where TP, TN, FP, and FN represent the counts of true positives, true negatives, false positives, and false negatives, respectively.

C. Compared With SOTA Methods

To highlight the advantages of the proposed FGPFormer, we make comparisons with the existing SOTA methods, which consist of FC-EF [8], FC-conv [8], FC-diff [8], ChangeFormer [12], BIT [13], SNUNet [10], DDRL [23], SFEARNet [11], and MaskCD [45]. All competing methods were evaluated strictly following their original experimental protocols, including data preprocessing, hyperparameter settings, and the number of training epochs. We compare against canonical FCN Siamese baselines that differ only in temporal fusion: FC-EF concatenates the bitemporal images at the input, while FC-conv and FC-diff employ weight-sharing Siamese encoders and merge multilevel features via concatenation or absolute differencing to highlight change cues. On top of CNN encoders, SNUNet enhances localization through densely connected skip pathways and attention-guided deep supervision, whereas SFEARNet targets efficiency and boundary fidelity

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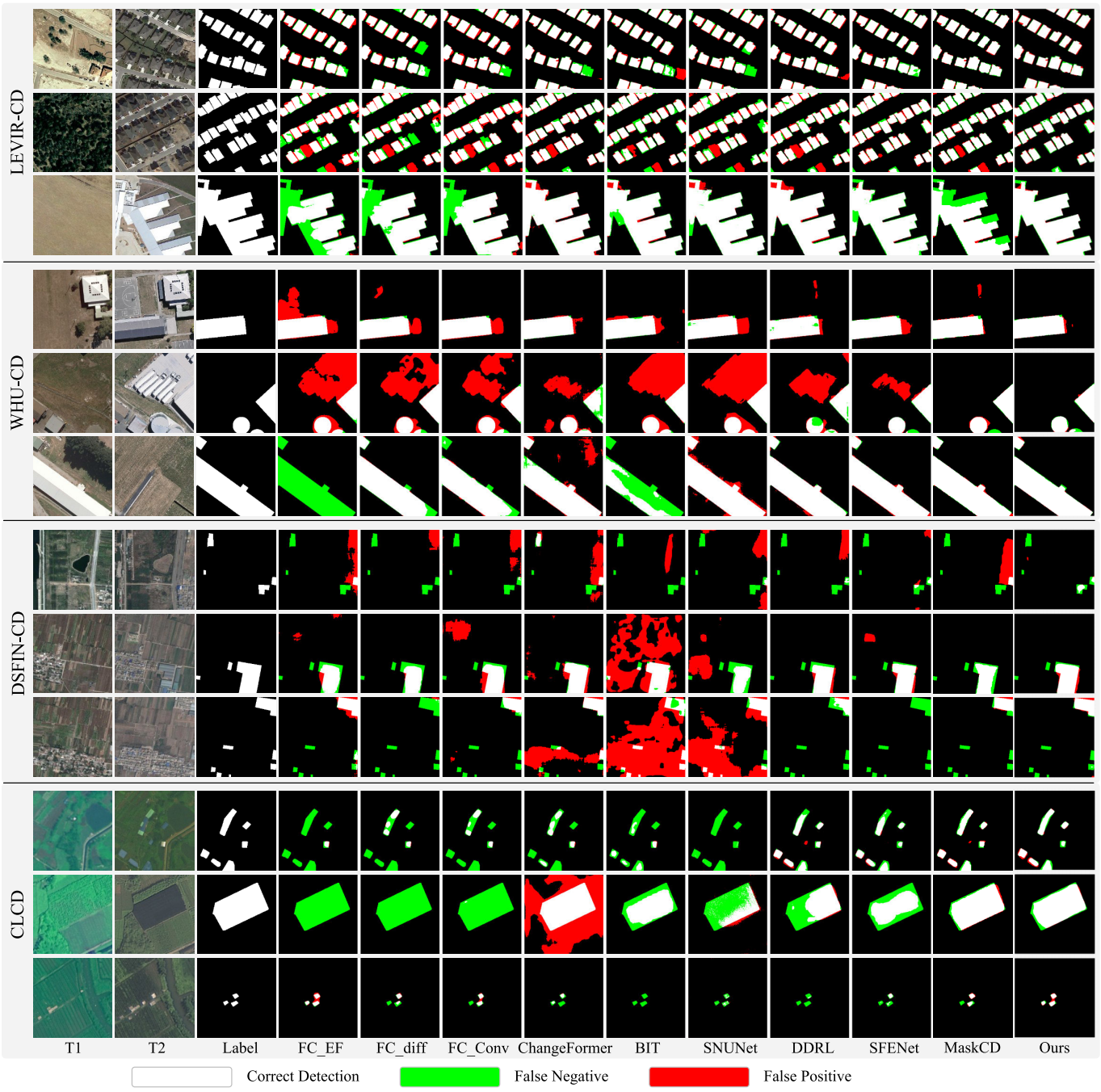


Fig. 4. Overall qualitative results of the proposed FGPFormer compared with SOTA methods on utilizing four datasets. White pixels indicate correct detections, green pixels indicate false negatives, and red pixels indicate false positives.

by coupling differential feature enhancement with semantic-flow transmission and an edge-aware refinement decoder. For global context modeling, BIT and ChangeFormer leverage Transformer-based interactions, which tokenize semantic context or hierarchical multiscale difference learning, to capture long-range dependencies, and DDRL further regularizes siamese Transformer features via distribution reconstruction and contrastive constraints to alleviate pseudo-changes.

1) Qualitative Evaluation: Qualitative evaluation on four benchmarks as shown in Figs. 4–7. In all qualitative results, white pixels indicate correct detections, green pixels indicate false negatives, and red pixels indicate false positives. Fig. 4 summarizes overall behavior across diverse scenes and highlights a common failure mode of prior methods, which indicate pseudo-change induced false alarms that spread over large homogeneous regions and boundary incompleteness around real changes. In contrast, the proposed FGPFormer consistently yields compact and semantically faithful masks with cleaner boundaries, indicating that frequency-guided restraint of appearance shift and progressive guidance injection effectively improve both precision and structural integrity. Fig. 5 focuses on fine-grained changes, where thin structures

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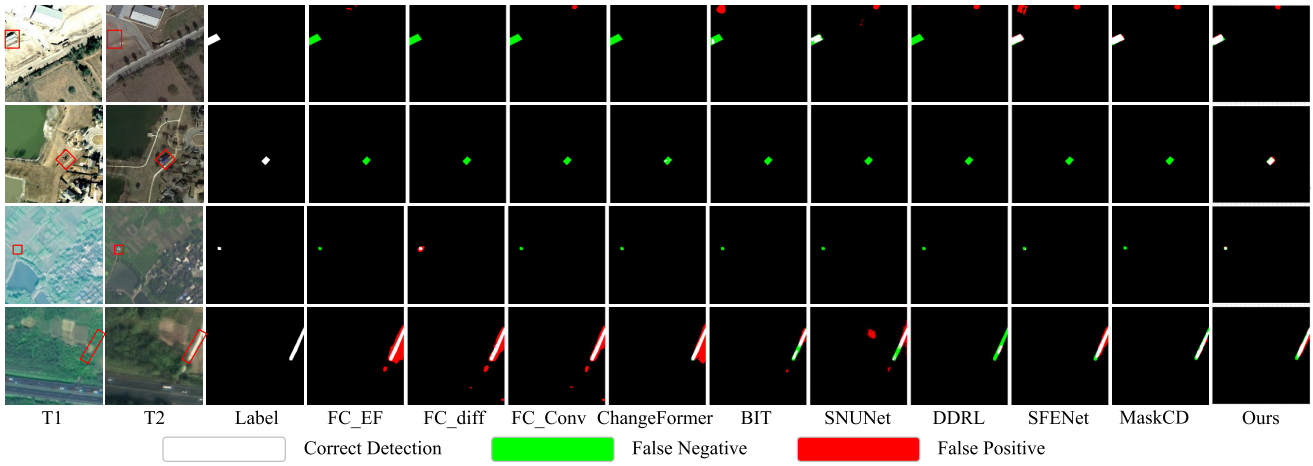


Fig. 5. Qualitative results in fine-grained changes of the proposed FGPFormer compared with SOTA methods on utilizing four datasets. White pixels indicate correct detections, green pixels indicate false negatives, and red pixels indicate false positives.

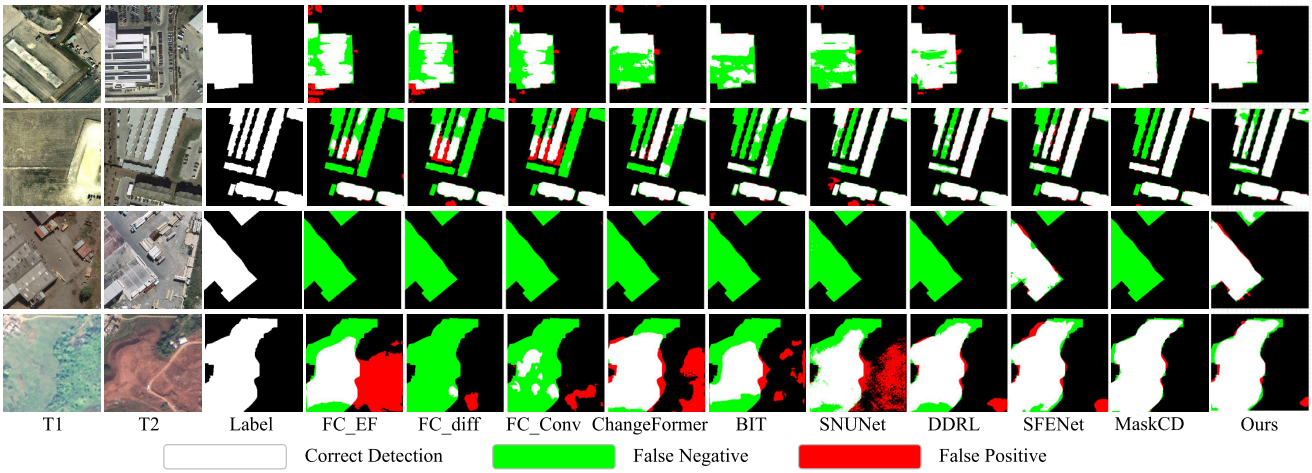


Fig. 6. Qualitative results in large-scale changes of the proposed FGPFormer compared with SOTA methods on utilizing four datasets. White pixels indicate correct detections, green pixels indicate false negatives, and red pixels indicate false positives.

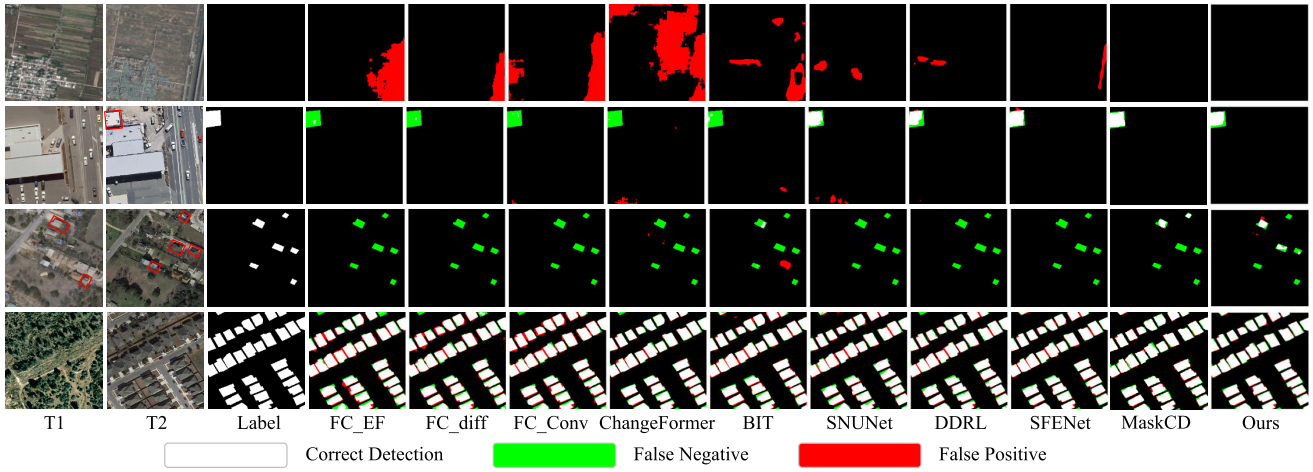


Fig. 7. Qualitative results in a complex background of the proposed FGPFormer compared with SOTA methods on utilizing four datasets. White pixels indicate correct detections, green pixels indicate false negatives, and red pixels indicate false positives.

and small objects are easily overwhelmed by low-frequency style shifts and by downsampling in the decoder. Competing methods often miss sparse changes or fragment elongated targets, whereas FGPFormer preserves continuity and recovers

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TABLE I
EXPERIMENTAL METRICS FOR THE FOUR UTILIZED DATASETS. THE BEST RESULTS ARE IN BOLD

Method	LEVIR-CD					WHU-CD					CLCD					DSIFN-CD				
	OA	P	R	F1	IoU	OA	P	R	F1	IoU	OA	P	R	F1	IoU	OA	P	R	F1	IoU
FC-EF [8]	98.10	82.95	78.95	80.90	67.93	98.06	75.31	76.02	75.66	60.85	93.75	58.40	55.40	56.86	39.72	78.75	43.16	79.18	55.87	38.76
FC-conv [8]	98.45	84.91	84.72	84.82	73.64	98.68	83.01	83.81	83.41	71.54	93.53	56.22	59.05	57.60	40.45	80.75	46.18	**80.26**	58.63	41.47
FC-diff [8]	98.44	87.07	81.41	84.15	72.63	98.74	85.05	82.71	83.86	72.21	93.39	56.23	50.27	53.09	36.13	87.8	66.09	57.89	61.72	44.63
ChangeFormer [12]	98.94	91.27	87.66	89.43	80.88	99.04	90.98	84.27	87.50	77.77	94.22	61.45	59.89	60.66	43.53	82.07	48.16	72.12	57.75	40.6
BIT [13]	98.80	89.54	86.45	87.97	78.52	97.25	62.02	78.89	69.45	53.20	94.22	61.39	60.31	60.82	43.73	77.84	41.1	70.25	51.86	35.01
SNUNet [10]	98.70	90.09	83.73	86.80	76.67	98.87	82.39	90.98	86.47	76.16	94.17	60.44	62.81	61.60	44.51	80.96	45.81	65.78	54.01	36.99
DDRL [23]	98.90	90.68	87.38	89.00	80.19	99.19	90.76	88.70	89.72	81.35	96.91	82.28	74.69	78.30	64.34	87.78	64.59	62.11	63.33	46.34
SFEARNet [11]	99.07	91.79	89.73	90.75	83.06	99.50	95.17	92.19	93.66	88.07	96.96	**82.61**	74.92	78.58	64.72	88.42	64.62	70.47	67.41	50.85
MaskCD [45]	99.03	91.65	89.12	90.36	82.42	99.53	95.64	**92.26**	93.92	**88.53**	96.61	81.77	75.27	78.38	64.46	86.39	58.05	71.70	64.16	47.23
Ours	**99.09**	**92.17**	**89.78**	**90.96**	**83.41**	**99.53**	**96.08**	91.78	**93.88**	88.46	**96.97**	82.09	**75.86**	**78.85**	**65.09**	**89.62**	**69.50**	69.32	**69.41**	**53.15**

TABLE II
EXPERIMENTAL METRICS OF ABLATION STUDY ON THE LEVIR-CD AND WHU-CD DATASETS. THE BEST RESULTS ARE IN BOLD

Backbone	TFA		SFR		CAS	LEVIR-CD					WHU-CD				
	AFS	TFG	DWT	MFA		OA	P	R	F1	IoU	OA	P	R	F1	IoU
MiT-b0						98.95	92.56	86.04	89.37	80.79	99.45	96.48	89.48	92.84	86.64
					✓	98.97	92.62	88.39	90.46	81.64	99.46	96.39	89.92	93.04	86.91
	✓				✓	99.08	92.56	89.03	90.76	83.08	99.47	96.09	90.67	93.30	86.86
		✓			✓	99.06	92.90	88.30	90.54	82.71	99.46	95.87	90.56	93.14	86.74
			✓		✓	99.08	92.92	88.62	90.72	83.01	99.52	**96.74**	90.19	93.35	87.02
				✓	✓	99.07	92.56	88.89	90.69	82.96	99.50	96.72	89.62	93.03	86.96
	✓	✓			✓	99.09	92.17	**89.78**	90.96	83.41	99.51	95.85	90.57	93.14	87.15
			✓	✓	✓	99.08	**92.97**	88.95	90.73	83.03	99.50	94.7	**91.95**	93.30	87.45
	✓	✓	✓	✓	✓	**99.10**	92.68	89.38	**91.00**	**83.49**	**99.53**	96.08	91.78	**93.88**	**88.46**

weak change evidence with fewer omissions. Fig. 6 reports large-scale changes, for which several baselines either leave holes inside large objects or produce jagged contours due to cross-scale inconsistency during decoding. FGPFormer produces more complete interiors and smoother outlines, suggesting that the learned frequency-structure guidance stabilizes region-level decisions while sharpening boundaries. Fig. 7 further considers complex backgrounds with strong texture confusion and nonchange distractors such as shadows and land-cover variations. Under these conditions, many methods exhibit elevated false positives in cluttered areas, while FGPFormer better separates real changes from background-induced differences and maintains coherent object shapes.

2) Quantitative Evaluation: Table I summarizes the quantitative comparison on LEVIR-CD, WHU-CD, DSIFN-CD, and CLCD under OA, precision, recall, $F1$, and IoU. FGPFormer consistently delivers the best OA-quality tradeoff across all benchmarks, achieving the highest $F1$ and IoU on every dataset. On the LEVIR-CD dataset, it improves $F1$ from 90.75% to 90.9% and IoU from 83.06% to 83.41% compared with the second best values, indicating more complete structure recovery with fewer fragmented detections. On the WHU-CD dataset, it further advances $F1$ of 93.88% and IoU of 88.46%, reflecting more faithful boundary delineation under dense urban textures. The advantage becomes more pronounced on the challenging DSIFN-CD and CLCD benchmarks, where pseudo-changes and background clutter are dominant. In particular, FGPFormer reaches $F1$ of 69.41% and IoU of 53.15% on the DSIFN-CD dataset, and $F1$ of 78.85% and IoU of 65.09% on the CLCD dataset, substantially exceeding the SOTA methods and demonstrating stronger robustness to shift-dominated activations while preserving high-frequency structural continuity.

TABLE III
EXPERIMENTAL METRICS OF TWO DIFFERENT FUSION STRATEGIES. THE BEST RESULTS ARE IN BOLD

Fusion Strategy	LEVIR-CD		WHU-CD		CLCD		DSIFN-CD	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU
Multiply	90.62	82.25	92.18	85.49	73.45	58.04	66.84	50.52
Sum	**90.96**	**83.41**	**93.88**	**88.46**	**78.85**	**65.09**	**69.41**	**53.15**

D. Ablation Study

In our ablation study, Table II validates the effectiveness and complementarity of each module on LEVIR-CD and WHU-CD datasets. This module consists of AFS and TFG of TFA, DWT, and MFA in SFR, and CAS. Starting from the baseline, introducing the CAS module improves LEVIR-CD from $F1$ of 89.37% and 80.79% of IoU to 90.46% and 81.64%, respectively, suggesting that change-aware interaction effectively strengthens discriminative evidence. Combining with CAS, the overall performance has been further enhanced by utilizing AFS, TFG, DWT, and MFA independently, indicating structural completeness and sharper change extent estimation with frequency guidance. Under the global frequency domain and direction feature-guided cross-attention of TFA, the $F1$ and IoU achieved 90.96% and 83.41%. Further enabling the full

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TABLE IV
EXPERIMENTAL METRICS FOR THE DSIFN AND DSIFN-RECONSTRUCTED DATASETS. THE BEST RESULTS ARE IN BOLD

Method	DSIFN-CD					DSIFN-CD-reconstructed				
	OA	P	R	F1	IoU	OA	P	R	F1	IoU
FC-EF [8]	78.75	43.16	79.18	55.87	38.76	82.85	69.23	74.37	71.71	55.89
FC-conv [8]	80.75	46.18	**80.26**	58.63	41.47	84.86	71.92	79.06	75.32	60.42
FC-diff [8]	87.8	66.09	57.89	61.72	44.63	85.40	82.34	63.67	71.81	56.02
ChangeFormer [12]	82.07	48.16	72.12	57.75	40.6	95.28	90.15	94.14	92.1	85.36
BIT [13]	77.84	41.1	70.25	51.86	35.01	89.61	79.88	86.16	82.9	70.79
SNUNet [10]	80.96	45.81	65.78	54.01	36.99	91.29	82.31	89.40	85.71	74.99
DDRL [23]	87.78	64.59	62.11	63.33	46.34	96.9	95.12	94.23	94.67	89.88
SFEARNet [11]	88.42	64.62	70.47	67.41	50.85	97.91	**95.95**	96.92	96.44	93.12
MaskCD [45]	87.33	60.19	75.14	66.84	50.20	97.41	94.78	96.46	95.61	91.59
Ours	**89.62**	**69.50**	69.32	**69.41**	**53.15**	**97.93**	95.59	**97.40**	**96.49**	**93.21**

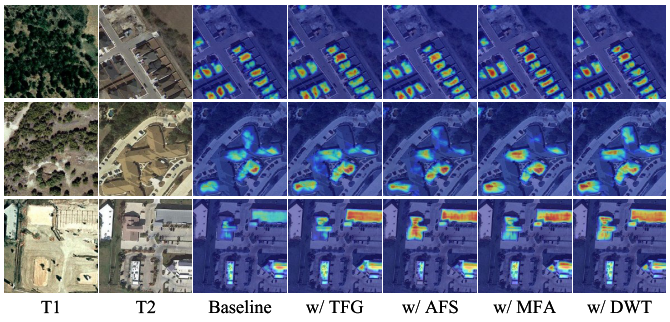


Fig. 8. Qualitative results of the ablation study of the TFA and SFR module on selected scenes. Grad-CAM is employed to measure the activation of different modules in response to the changed regions.

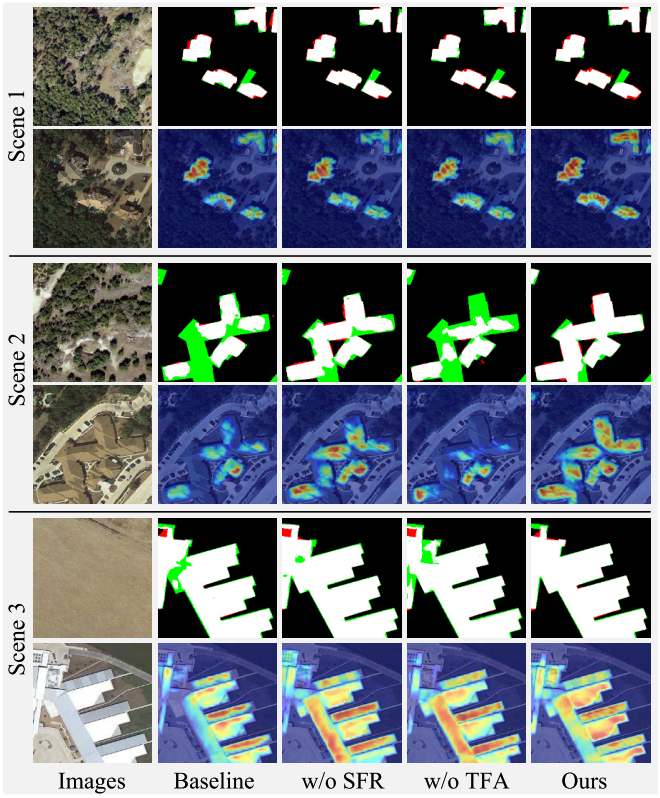


Fig. 9. Qualitative results of the ablation study of the proposed FGPFormer on selected scenes. White pixels indicate correct detections, green pixels indicate false negatives, and red pixels indicate false positives. Grad-CAM [46] is employed to measure the activation of different modules in response to the changed regions.

frequency interaction branch raises IoU to 83.08, indicating better structural completeness and sharper change extent estimation. Incorporating the structural refinement module yields consistent gains on both datasets, which is reflected by the progressive increase of IoU and the more balanced precision-recall behavior. Finally, coupling structural refinement with the proposed calibration strategy produces the best results, reaching $F1$ of 91.00% and IoU of 83.49% on LEVIR-CD and $F1$ of 93.88% and IoU of 88.46% on WHU-CD. Overall, the ablation results show that removing TFA leads to more pseudo-change responses and false alarms, while removing SFR mainly degrades boundary quality and structural completeness. The full model achieves the best balance between suppressing spurious activations and preserving object-level geometry. Moreover, the results presented in Table III further confirm that additive fusion serves as the optimal integration strategy for both TFA and SFR within the CAS, exhibiting superior effectiveness and better suitability compared with the multiplicative fusion strategy.

We conduct qualitative ablations to isolate the contribution of TFA and SFR, with representative scenes shown in Figs. 8 and 9. Fig. 8 provides a stepwise analysis of key design components inside TFA and SFR, which include AFS, TFG, DWT, and MFA. First, directional texture guidance improves the continuity of anisotropic changes and alleviates discontinuities along linear structures, and Fourier frequency selection further reduces low-frequency shift, thereby decreasing over-detection in smooth backgrounds. Subsequently, the subsequent feature fusion stage integrates complementary evidence and yields more stable localization across different object scales. Finally, wavelet-based refinement strengthens edge sharpness and reduces interior holes, producing the most consistent and visually plausible change masks among the ablated variants. Fig. 9 compares the baseline model, the variant without SFR, the variant without TFA, and the proposed FGPFormer. Baseline indicates SegFormer [47] with shared MiT-b0 backbone for bitemporal RS images. Removing TFA substantially increases pseudo-change responses, which manifest as widespread false alarms in visually similar but unchanged regions. Removing SFR primarily degrades boundary quality and structural completeness, leading to fragmented

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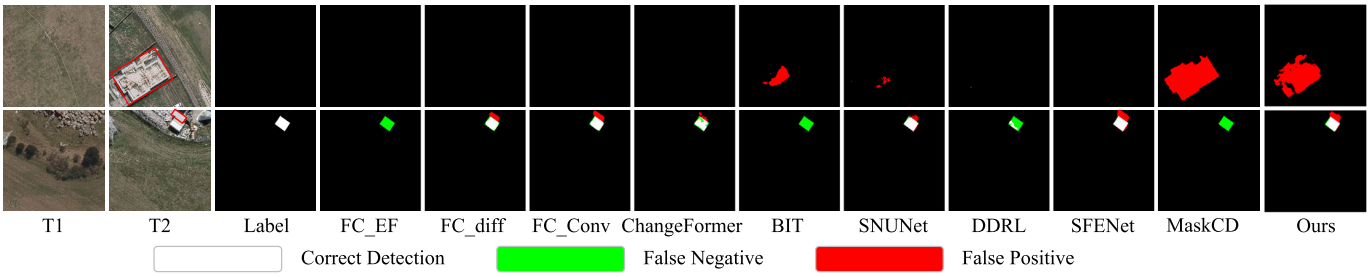


Fig. 10. Qualitative results on annotation defects of the proposed FGPFormer compared with SOTA methods. The red box indicates the missing labeled regions. White pixels indicate correct detections, green pixels indicate false negatives, and red pixels indicate false positives.

TABLE V
EXPERIMENTAL METRICS OF COMPUTATIONAL EFFICIENCY ON THE CLCD DATASET. THE BEST RESULTS ARE IN BOLD

Method	Params(M)	FLOPS(G)	FPS
FC-EF [8]	1.35	7.12	487.83
FC-conv [8]	1.55	10.6	349.13
FC-diff [8]	1.35	9.39	342.11
ChangeFormer [12]	41.03	202.79	44.88
BIT [13]	3.5	21.27	89.70
SNUNet [10]	12.03	109.67	124.45
DDRL [23]	33.55	42.55	18.83
SFEARNet [11]	5.56	4.65	25.11
MaskCD [45]	107.41	138.67	6.57
Ours	10.01	11.92	33.38

masks, broken contours, and missed thin components. The proposed FGPFormer achieves the best balance, simultaneously suppressing spurious activations and refining object-level geometry.

E. Discussion

1) Generalization Analysis: Table IV demonstrates generalization under different distribution reconstructions on DSIFN-CD dataset. We constructed DSIFN-CD-reconstructed dataset, which evaluates the overall performance based on DSIFN-CD dataset in Section IV-B2. On the one hand, DSIFN-CD dataset tests the cross-domain capabilities, and the proposed FGPFormer has achieved the SOTA performance in both qualitative and quantitative indicators. As shown in Fig 4, the proposed FGPFormer has obtained precise RSCD in complex small targets and buildings with clear structural information. On the other hand, FGPFormer acquired the best results in all five evaluation metrics we utilized under the comprehensive evaluation of the DSIFN-reconstructed dataset. Specifically, when transferring from DSIFN-CD to the reconstructed setting, FGPFormer not only preserves performance but also improves it, increasing $F1$ from 69.41% to 96.49% and IoU from 53.15% to 93.21%. The simultaneous rise in both recall and precision indicates that the model maintains stable sensitivity to true structural changes under appearance shifts, which supports the claim that frequency-guided interaction mitigates shift-sensitive activations and improves robustness beyond the original training distribution.

2) Complexity Analysis: As shown in Table V, computational efficiency is analyzed in terms of parameter counts (Params), floating point operations (FLOPs), and frames per second (FPS) of the inference procedure. Params and FLOPs are calculated employing a tensor in size of $1 \times 3 \times 256 \times 256$, and FPS is statistically obtained in the inference process of the CLCD dataset. The proposed FGPFormer contains 10.01M parameters and requires 11.92G FLOPs, which remains close to lightweight convolutional baselines while being substantially more economical than attention-heavy architectures. For example, ChangeFormer introduces 41.03M parameters and 202.79G FLOPs, and SNUNet requires 109.67G FLOPs, indicating that our frequency-guided progressive design avoids the excessive arithmetic cost often incurred by dense decoders or global interaction modules. Although FGPFormer is hard to exhibit significant advantages over SFEARNet in Params and FLOPs, in actual reasoning, FPS is approximately 33% higher than SFEARNet. Overall, these results suggest that FGPFormer delivers strong accuracy improvements with a controlled computational footprint, making it suitable for large-scale and high-resolution change monitoring.

TABLE VI
INFERENCE FPS AND PEAK GPU MEMORY CONSUMPTION UNDER DIFFERENT INPUT PATCH SIZES ON THE CLCD DATASET

Patch Size	FPS	Peak Memory(GB)
256×256	33.38	0.30
512×512	29.38	0.71
768×768	19.45	1.36
1024×1024	10.81	2.38

Moreover, Table VI demonstrates the excellent memory efficiency of our method: the peak memory for 1024×1024 input is only 2.38 GB, which readily supports patch-wise sliding inference for ultralarge RSIs. Inference speed degrades moderately with rising resolution, retaining 10.81 FPS at 1024×1024. Benefiting from the lightweight design, it well satisfies the efficiency demands of practical large-scale remote sensing monitoring.

3) Annotation Defects: In our experiments, a number of annotation defects in the annotations are identified, whether they are analyzed from the experimental results or discovered manually. These defects mainly involve the omission of marking significant true changes. As illustrated in Fig. 10, it highlights annotation errors that complicate both training and evaluation. Boundary imprecision, missing labels for small objects, and inconsistent inclusion criteria across annotators

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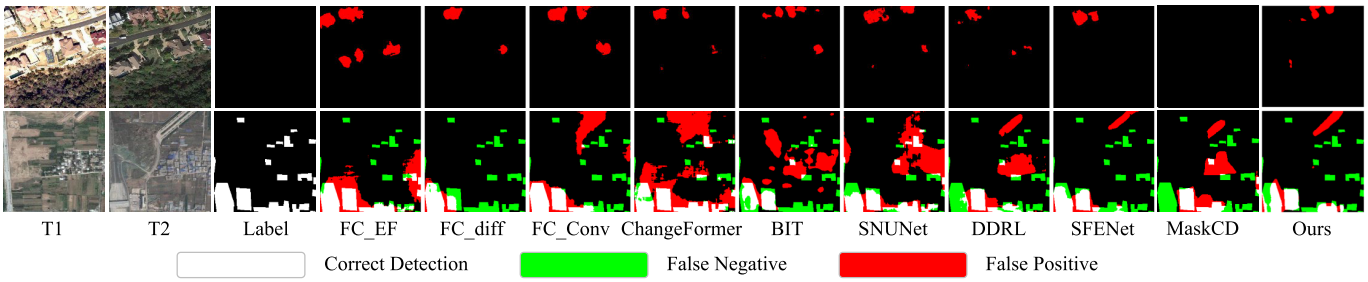


Fig. 11. Qualitative results in limitations of the proposed FGPFormer and other SOTA methods. White pixels indicate correct detections, green pixels indicate false negatives, and red pixels indicate false positives.

can penalize structurally reasonable predictions as false positives or false negatives, particularly near object edges and in fine-grained regions. This phenomenon also partially explains why qualitative improvements may appear more significant than what pixel-level metrics reflect when labels are noisy. On the other hand, although the outcome of the proposed FGPFormer is indicated as a false positive in the qualitative results, in actual situations, its detection is correct, which demonstrates that FGPFormer has significant anti-noise and robustness performance.

4) Limitations: As shown in Fig. 11, extreme radiometric discrepancies still blur the distinction between subtle structural changes and nonsemantic variations, especially for spatially thin targets under heavy background clutter. Future work investigates explicit alignment modeling and stronger radiometric invariances to further mitigate misregistration and appearance shift in challenging deployments. In addition, although transformer-based models typically achieve superior recognition accuracy due to long-range dependency modeling as demonstrated in Table V, their attention computation scales quadratically with the number of tokens, which could become a bottleneck for very high-resolution RSIs. This limitation demands for further exploration of lightweight transformer design, including token-efficient interaction, locality-aware attention, and compression strategies that preserve boundary fidelity while reducing the quadratic cost.

V. Conclusion

In this work, we presented FGPFormer, a frequency-guided progressive transformer for remote sensing CD that explicitly disentangles appearance shift-driven pseudo-changes from structure-driven true changes via coupled frequency interaction and structure-aware refinement. The TFA module performs adaptive Fourier feature interaction to attenuate low-frequency shift and enhance discriminative change evidence, while the SFR module leverages multiband wavelet cues to strengthen connectivity and sharpen boundaries, and the calibration strategy further balances precision and recall. Extensive experiments on four public benchmarks demonstrate consistent SOTA performance. In future work, we will focus on noise-aware supervision and uncertainty modeling, as well as explicit alignment and geometry reasoning to further mitigate misregistration and strengthen invariance to radiometric distortions.

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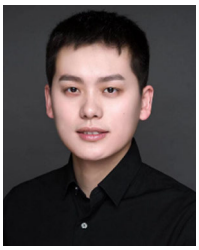
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